Sahaj Bhadja

(213) 284-4651 | bhadja@usc.edu | LinkedIn | GitHub | Portfolio

Summary

Software engineer with 2+ years building secure web applications, APIs, distributed workflows, and production services. Experienced in Java, Python, JavaScript, TypeScript, automated testing, CI/CD, access controls, debugging, observability, and reliable software delivery.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built Python APIs, Azure SQL services, and secure dashboards for 500+ users, tuning indexed queries and backend access paths to reduce production response latency from 1.2 seconds to 350 milliseconds.
- Implemented reliable stateful workflows for 5,000+ monthly jobs using explicit transitions, retries, reconciliation, and failure isolation, tripling throughput while improving correctness, observability, maintainability, and production issue diagnosis.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed React interfaces and Flask REST services handling 10,000+ daily requests, implementing secure file ingestion, authentication, status tracking, and grading workflows while resolving production issues for 150+ users.
- Created concurrent processing for 1,000+ files with automated validation, AWS RDS persistence, structured logging, and failure handling, reducing turnaround 60% while maintaining reliable application behavior under production load.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed Python services, SQL integrations, and ETL workflows using AWS Glue, S3, and REST APIs, improving response times 30% and data-processing efficiency 68% across secure production applications.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito across Windows, Linux, and macOS, accelerating defect diagnosis, code review feedback, and reliable production releases.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular React and Node.js features across eight services with JWT authentication and role-based access controls, synchronizing 2 million+ Azure SQL records through maintainable APIs and real-time WebSockets.
- Automated Docker builds, tests, staged deployments, and production monitoring through Jenkins and Azure, reducing release time 50% while improving code quality, deployment consistency, and operational visibility across environments.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built Flask APIs and responsive workflows for reservations, status updates, and payments across 50,000+ records, translating operational requirements into maintainable features with validation and secure Firebase access.
- Implemented Firebase persistence, webhooks, and payment synchronization with structured validation and logging, reducing manual payment handling 70% while improving issue triage, transaction reliability, and responsive user support.

Skills

Languages & Web: Java, Python, JavaScript, TypeScript, C++, SQL, React, Next.js, Node.js, HTML5, CSS3, responsive interfaces
Backend & Security: Flask, FastAPI, REST APIs, microservices, WebSockets, authentication, authorization, JWT, role-based access control, secure file ingestion, API integration
Data & Cloud: PostgreSQL, Azure SQL, SQL Server, MySQL, MongoDB, Firebase, AWS RDS, AWS, Azure, S3, Blob Storage, data modeling, query optimization
Quality & Operations: Git, Docker, Kubernetes, Jenkins, CI/CD, PyTest, Mockito, unit testing, integration testing, code review, debugging, logging, monitoring, observability, production support, Agile

Projects

Chat2SQL | Code | *Java, TypeScript, SQL, Maven, Docker, REST APIs*
- Built a Java and TypeScript developer tool supporting natural-language queries across PostgreSQL, MySQL, and MongoDB, with Docker packaging, REST APIs, validation, and reproducible Maven-based builds for reliable delivery.

FinNexus | Live | *TypeScript, React, REST APIs, WebSockets, MongoDB, Kubernetes*
- Built secure financial workflows supporting 50,000+ users and one million transactions, integrating REST APIs, real-time synchronization, granular filters, MongoDB persistence, monitoring, validation, and automated Kubernetes production deployment.

Big-Brain | Live | *LLaMA-2, FAISS, PostgreSQL, REST APIs, Clerk, Convex*
- Built an authenticated document platform using LLaMA-2, FAISS, PostgreSQL, and REST APIs, enforcing Clerk-based access controls while supporting conversational search and real-time collaboration across 50,000+ user documents.